THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A planet has a thick CO2 atmosphere and a surface hot enough to melt lead. This planet is most like: 1

(a) Earth **(b)** the Moon **(c)** Mars **(d)** Venus

2. The thin air at the top of Mount Everest has about one-third of the air pressure at sea level. This means a breath of air contains: 1

(a) the same oxygen **(b)** about one-third the oxygen
(c) no oxygen **(d)** three times the oxygen

3. Assertion (A): Most of the Earth's water is not directly available for drinking. Reason (R): Most water is salty or frozen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. If the Earth had no atmosphere, the day-night temperature difference would: 1

(a) decrease **(b)** vanish **(c)** remain same **(d)** increase greatly

5. A thermometer in the thermosphere reads thousands of degrees but we would not feel hot because: 1

(a) ozone protects us
(b) it is humid
(c) there are very few particles to transfer heat
(d) it is dark

6. Assertion (A): The Earth's magnetic field is important for life. Reason (R): It deflects the harmful charged particles from the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Which instrument helps in knowing about earthquake waves? 1

(a) barometer **(b)** seismograph **(c)** anemometer **(d)** thermometer

8. Acid rain forms from the oxides of sulphur and nitrogen, which come mostly from: 1

(a) burning fossil fuels **(b)** forests
(c) oceans **(d)** volcanoes only

9. Assertion (A): Deforestation contributes to global warming. Reason (R): Trees absorb carbon dioxide during photosynthesis. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. When coal is burnt in power plants, which of the following increases in the atmosphere? 1

(a) nitrogen **(b)** argon **(c)** carbon dioxide **(d)** helium

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we not feel the pressure of the atmosphere on our bodies? 2

12. Why is the ozone layer thinner at the poles in spring? 2

13. Why is the greenhouse effect a necessary natural process, but its enhancement is harmful? 2

14. A town gets 600 mm of rain yearly over 10 square km. Find the volume of rain in cubic metres (1 mm over 1 square m is 1 litre). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the average global temperature is expected to rise by a few degrees if CO_2 doubles and give three consequences. 3

16. Why do we call water 'a renewable but limited resource'? 3

17. Explain the impact of the pollution of the troposphere in cities and how it differs from stratospheric changes. 3

OR

Explain how oceans moderate climate and absorb CO_2. 3

18. Explain how forests, wetlands and oceans act as 'sinks' for CO_2. 3

OR

Why are glaciers called 'water towers' of the world? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain, with reference to each sphere, how a volcanic eruption affects the Earth system in the short and long term. 5

OR

Explain why the shell of life (biosphere) is extremely thin compared with the Earth's size, using the data of radius 6400 km and typical biosphere thickness of about 20 km. 5

20. Design a school project on the 'carbon footprint' of the school: collecting data, calculating, analysing and recommending three actions. 5

OR

Discuss the water crisis in India and measures like rainwater harvesting, watershed management, drip irrigation and reuse of wastewater, with examples including Rajasthan's traditional methods such as johads and baoris. 5

END OF QUESTION PAPER — **Total: 40 Marks**